

Review of Algebra and Trigonometry

Section 1.2

A. Functions and Relations

DEFINITION Relation: A set of ordered pairs.
 $(x, y) \rightarrow (\text{domain}, \text{range})$

DEFINITION Function: A correspondence from one set (the domain) to another set (the range) such that each element in the domain corresponds to exactly one element in the range.

Example: Determine whether each of the following is an example of a function or not.

1. $\{(1,1)(3,2)(5,3)\}$

2. $\{(1,1)(2,4)(3,-5)(2,-4)\}$

3. $x^2 + 2y = 0$

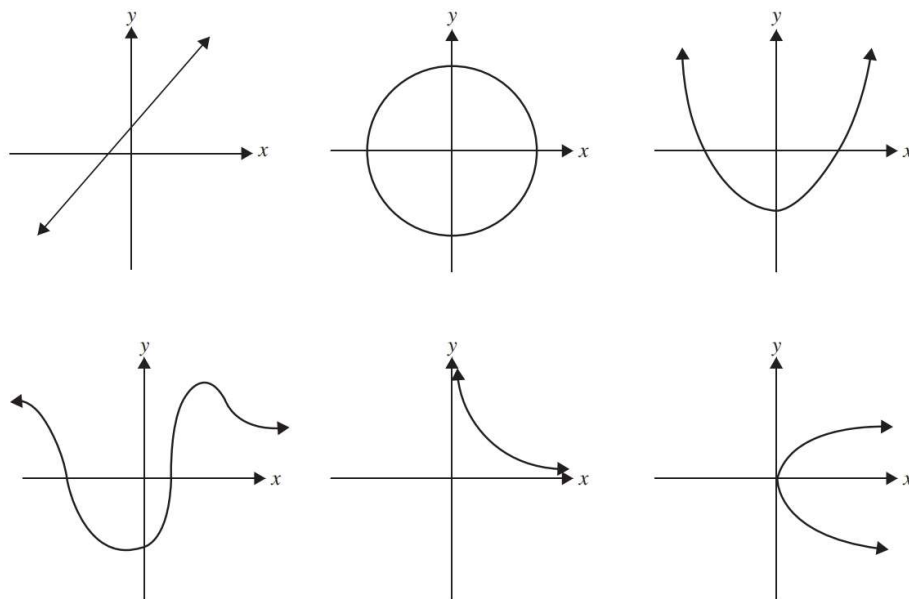
4. $x^2 + y^2 = 1$

Vertical Line Test for Functions

Vertical Line Test for Functions: If any vertical line intersects a graph more than once, then the graph is **not** a function.

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Example: Determine whether each of the following is a function or not by the Vertical Line Test.



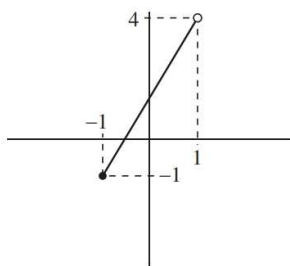
B. Domain and Range of a Function

DEFINITION Domain: Input $\rightarrow x$ values (i.e., all of the values of x that I may plug into a function)

DEFINITION Range: Output $\rightarrow y$ values (i.e., all of the values of y that a function can attain)

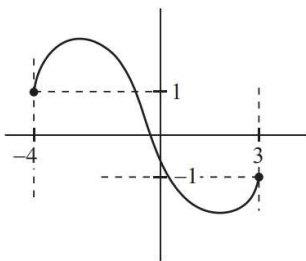
Example: Give the domain and range (in interval notation) for each of the following:

1.



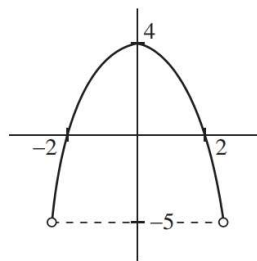
Domain: _____

Range: _____



Domain: _____

Range: _____



Domain: _____

Range: _____

2. $g(x) = \sqrt{2-x}$

3. $f(x) = \frac{x^2 - 4}{x + 2}$

4. $g(x) = \frac{3x}{\sqrt{x+5}}$

5. $h(x) = \ln(6x - 3)$

*The three functions for which we will most frequently have domain restrictions (in this course) are **fractions**, **radicals**, and **logarithms**.

C. Linear Models

Definition

$$\text{Slope} = m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope	$m = \frac{y_2 - y_1}{x_2 - x_1}$	(Use when given two points to find slope)
Slope–Intercept Form	$y = mx + b$	(Use when given slope and y intercept)
Point–Slope Form	$(y - y_0) = m \cdot (x - x_0)$	(Use when given one point and slope)
General Form	$A \cdot x + B \cdot y + C = 0$	
Horizontal Line	$y = b$ (where $b = \text{constant}$)	
Vertical Line	$x = c$ (where $c = \text{constant}$)	

Parallel Lines

Two lines are parallel if and only if they have the same slope.

For two lines $y_1 = m_1x + b_1$ and $y_2 = m_2x + b_2$ we have $y_1 \parallel y_2 \Leftrightarrow m_1 = m_2$

Perpendicular lines

Two lines are perpendicular if and only if the product of their slope $= -1$.
For two lines $y_1 = m_1x + b_1$ and $y_2 = m_2x + b_2$ we have $y_1 \perp y_2 \Leftrightarrow m_1 = -\frac{1}{m_2}$

Examples: Find the equation of the line:

1. That passes through point $(0, -3)$ with slope $= -2$
2. That passes through points $(3, -2)$ and $(4, 5)$
3. That passes through point $(0, 0)$ and is parallel to the line $y = 3x - 9$
4. That passes through point $(2, -4)$ and is perpendicular to the line $y = \frac{1}{2}x + 3$
5. Find the slope and y intercept of the line $9x - 3y - 3 = 0$

D. Classes of Functions

1. Power Functions

For any real number m , a function in the form $f(x) = x^m$ is called a Power Function

2. Polynomials

Definition

A *polynomial function* is a function in the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0$ where $a_0 \neq 0$ and n is a positive integer.

Examples: State whether each is a polynomial:

1. $g(x) = x^2 + 5x + 6$

2. $f(x) = x^3 + 7x - \sqrt{x}$

3. $h(x) = x^{\frac{2}{3}} - x + 5$

4. $f(x) = \frac{x^2}{3} + 8x$

3. Rational Functions

A Rational Function is the quotient of two polynomial functions:

A Rational Function is a function of the form $f(x) = \frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0}$

Asymptotes

An asymptote is an imaginary line that the graph of a function approaches as the function approaches a restricted number in the domain or as it approaches infinity.

1. Locating Vertical Asymptotes

If $f(x) = \frac{p(x)}{q(x)}$ is a rational function, $p(x)$ and $q(x)$ have no common factors and n is a zero of $q(x)$, then the line $x = n$ is a vertical asymptote of the graph of $f(x)$.

II. Locating Horizontal Asymptotes

$$\text{Let } f(x) = \frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0}$$

- i. If $n < m$, then $y = 0$ is the horizontal asymptote.
- ii. If $n = m$, then the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
- iii. If $n > m$, there is NO horizontal asymptote. (But there will be a slant/oblique asymptote.)

Examples: Find all asymptotes.

1. $f(x) = \frac{15x}{3x^2 + 1}$

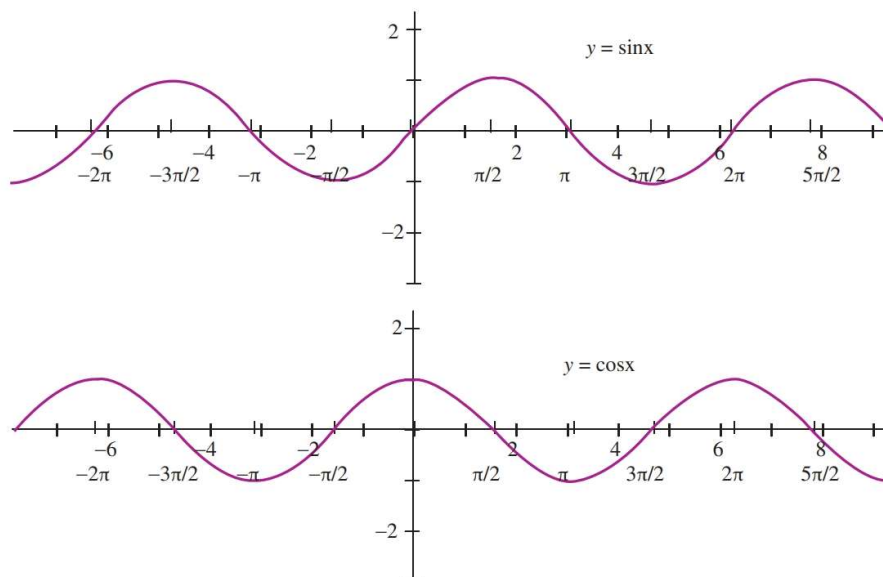
2. $g(x) = \frac{15x^3}{3x^2 + 1}$

3. $h(x) = \frac{-3x + 7}{5x - 2}$

4. $k(x) = \frac{2x - x^2}{x^2 - 2x - 3}$

4. Trigonometric Functions

$\sin x$	$\csc x$
$\cos x$	$\sec x$
$\tan x$	$\cot x$



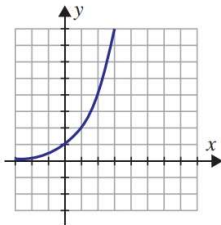
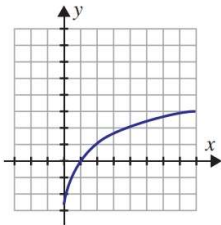
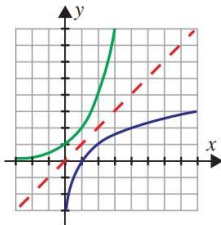
5. Exponential and Logarithmic Functions

DEFINITION: An exponential function is a function in the form $f(x) = a^x$. (i.e., the variable x is in the exponent).

DEFINITION: A logarithmic function is a function in the form $f(x) = \log_a x$. (i.e., the variable x is in the expression).

$y = \log_b x$ “ y is equal to log base b of x ”—Here “ b ” is the BASE NUMBER and “ x ” is the VARIABLE.

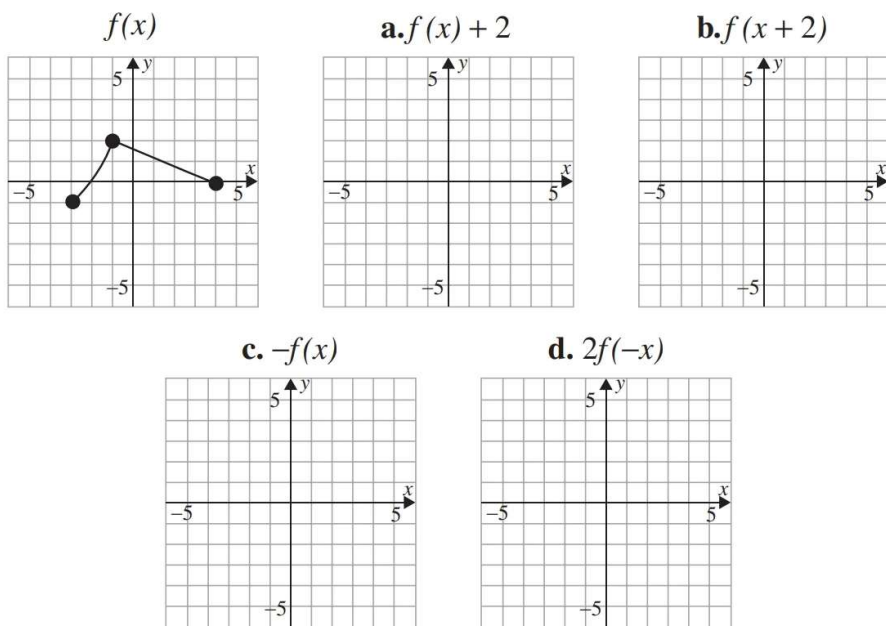
$\log_b x = y$ means exactly the same thing as $b^y = x$

$y = 2^x$	$y = \log_2(x)$	Comparison of the two graphs, showing the inversion line in red.
		

E. Transformations of Functions

Vertical Shifts	$f(x) + C$	↑	Moves graph UP C units
	$f(x) - C$	↓	Moves graph DOWN C units
Horizontal Shifts	$f(x - C)$	→	Moves graph RIGHT C units
	$f(x + C)$	←	Moves graph LEFT C units
Vertical and Horizontal Reflections	$-f(x)$	↕	Flips graph about x-axis
	$f(-x)$	↔	Flips graph about y-axis
Vertical Stretching/ Compressing	$c \cdot f(x)$ for $c > 1$	↑	Graph vertically stretches by a factor of C
	$c \cdot f(x)$ for $0 < c < 1$	↓	Graph vertically shrinks by a factor of C

Example: Use the given graph of $f(x)$ to sketch each of the following:

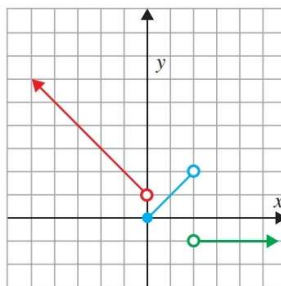


F. Combinations of Functions

1. Piecewise-Defined Functions

A Piecewise Function is a function that has specific (and different) definitions on specific intervals of x .

$$f(x) = \begin{cases} -x+1 & x < 0 \\ x & 0 \leq x < 2 \\ -1 & x > 2 \end{cases}$$



2. Sums, Differences, Products, and Quotients of Functions

Sum	$(f + g)(x) = f(x) + g(x)$
Difference	$(f - g)(x) = f(x) - g(x)$
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

3. Composition of Functions

$f(x) = x^2 + 5x + 2$	$f(\odot) = (\odot)^2 + 5(\odot) + 2$
$g(x) = \frac{2x}{\sqrt{x+1}}$	$g(\quad) = \frac{2(\quad)}{\sqrt{(\quad)+1}}$
$k(x) = 2x - 3$	$k(\quad) = 2(\quad) - 3$
$h(x) = \sqrt{x^2 + 5x}$	$h(\quad) = \sqrt{(\quad)^2 + 5(\quad)}$ $= \sqrt{(\quad)^2 + 5(\quad)}$
Notation	$(f \circ g)(x) = f(g(x))$

Example

1. For the functions $f(x) = \sqrt{x}$ and $g(x) = x + 2$ find

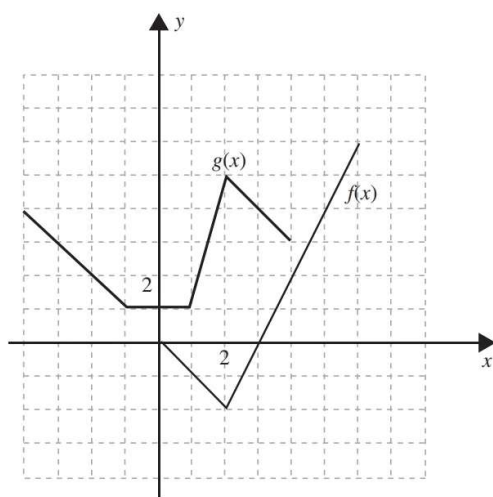
a. $(f \circ g)(x) =$

b. $(g \circ f)(x) =$

c. $(f \circ g)(2) =$

d. $(g \circ g)(x) =$

Example: For the functions $f(x)$ and $g(x)$ given in the graph find



a. $(f \circ g)(2) =$

b. $(f \circ g)(3) =$

c. $(g \circ f)(2) =$

d. $(g \circ f)(3) =$

e. $(f \circ f)(4) =$

f. $(g \circ g)(4) =$